

Comparative Planetary Magnetobiology of the Moon and Mars: Integrated Geophysical Mapping, Hypomagnetic Stress Synthesis, and Operational Life Support Frameworks for Deep-Space Exploration

Richard Barker^{1,*}, Adriana Kaley Sanchez¹, Manisha Dagar¹, Katrina Boland¹, Cauê Sciascia Borlina², D. Marshall Porterfield¹

¹Department of Agricultural and Biological Engineering, Purdue University, West Lafayette, IN 47907, USA

²Department of Earth, Atmospheric, and Planetary Sciences, Purdue University, West Lafayette, IN 47907, USA

*Corresponding author: rbarker2@purdue.edu

ABSTRACT

Terrestrial life evolved over four billion years within the continuous, protective presence of Earth's geomagnetic field (GMF; $\sim 50 \mu\text{T}$). As international space agencies and commercial ventures prepare for sustained human presence on the Moon (Artemis program) and Mars, biological systems—including crops in bioregenerative life support systems (BLSS), closed-loop microbiomes, and human crew—will be permanently uprooted from this evolutionary baseline and deployed into severe hypomagnetic field (HMF) environments. In this study, we present an integrated planetary magnetobiology investigation combining calibrated orbital and surface magnetometer datasets across 36 candidate, historical, and contemporary landing sites on the Moon (23 sites) and Mars (13 sites) with a comprehensive synthesis of terrestrial magnetobiology literature. We show that all 13 Artemis III candidate South Polar landing regions on the Moon reside within an ultra-low, near-null crustal magnetic baseline ($|\mathbf{B}_{30\text{km}}| < 1.0 \text{ nT}$, $> 50,000\times$ to $> 125,000\times$ weaker than Earth). On Mars, we resolve a profound hemispheric dichotomy: ice-rich northern lowland candidate habitat sites (Arcadia Planitia, Deuteronilus Mensae) and rover sites (Jezero Crater, Oxia Planum) experience extreme near-null hypomagnetism ($|\mathbf{B}_{\text{surf}}| < 25 \text{ nT}$), whereas southern highland anomaly belts (Terra Sirenum) harbor intense remnant crustal fields ($|\mathbf{B}_{\text{surf}}| > 1500 \text{ nT}$) capable of forming mini-magnetospheres. InSight magnetometry provides ground truth confirming local subsurface amplification ($\sim 2013 \text{ nT}$). We synthesize empirical biological evidence demonstrating that HMF exposure disrupts cryptochrome- and phytochrome-mediated light sensing, PIN2 polar auxin transport, iron homeostasis, and thylakoid ultrastructure in plants, while inducing circadian arrhythmicity, altered DNA repair kinetics, and accelerated osteoclastic bone demineralization in mammalian systems. Finally, we formulate a multi-planetary biological risk matrix and establish operational recommendations, emphasizing active Helmholtz coil magnetic restoration ($\sim 50 \mu\text{T}$) in surface growth modules and standardized in situ magnetometry on precursor landers.

Keywords: Comparative planetary magnetobiology, Hypomagnetic field (HMF), Artemis landing sites, Mars human habitats, Bioregenerative life support, Cryptochrome signaling, Space biology

1 Introduction

Terrestrial life evolved over nearly four billion years within the omnipresent, protective envelope of Earth's geomagnetic field (GMF), which delivers a continuous surface magnetic flux density ranging from $\sim 25 \mu\text{T}$ at the equator to $\sim 65 \mu\text{T}$ at the magnetic poles. On a planetary scale, the GMF serves as a primary shield against the solar wind and galactic cosmic radiation (GCR). On a cellular and physiological scale, the GMF acts as a pervasive environmental signal that has shaped fundamental biological mechanisms across all domains of life^{1,2}. From cryptochrome-mediated light sensing and radical pair spin recombination kinetics^{3,4} to polar auxin transport in plant root gravitropism⁵, circadian rhythm entrainment⁶, and cellular redox homeostasis⁷, biological systems are intrinsically calibrated to Earth's magnetic baseline.

As humanity transitions from low Earth orbit (LEO) to sustained multi-planetary exploration, space missions will permanently detach biological systems from this terrestrial magnetic

cradle. In deep space, on the lunar surface, and across Martian landscapes, organisms will experience chronic exposure to hypomagnetic fields (HMF)—defined as any magnetic environment with an intensity significantly below Earth's baseline ($< 5 \mu\text{T}$). While interplanetary transit exposes payloads to isotropic interplanetary magnetic fields (IMF) of $\sim 5 \text{ nT}$, surface operations on the Moon and Mars present complex, highly heterogeneous planetary magnetic environments resulting from the ancient extinction of their internal core dynamos^{8,9}.

The paleomagnetic histories of the Moon and Mars diverged dramatically from Earth's continuously active core dynamo:

1. **The Moon:** The early lunar core dynamo ceased operation between ~ 3.5 and 1.9 billion years ago⁹. Today, the Moon lacks any global dipole field, preserving only localized, weak crustal magnetic anomalies concentrated primarily in the far-side South Pole-Aitken (SPA) basin and impact basin antipodes^{10,11}.
2. **Mars:** Mars experienced a catastrophic core dynamo shut-

down approximately 4.1 to 3.9 billion years ago during the late Noachian era^{8,12}. Unlike the Moon, Mars retains extremely intense, large-scale remnant crustal magnetization embedded within its ancient southern highland crust¹³. This creates a profound planetary dichotomy between demagnetized northern lowlands ($|\mathbf{B}_{\text{surf}}| < 25 \text{ nT}$) and intensely magnetized southern anomaly belts ($|\mathbf{B}_{\text{surf}}| > 1500 \text{ nT}$) capable of generating mini-magnetospheres^{14,15}.

Despite extensive space biology research into microgravity and space radiation, the physiological consequences of planetary magnetic field heterogeneity have remained largely unexamined in integrated mission planning. Controlled terrestrial HMF simulations reveal that magnetic deprivation induces significant molecular and phenotypic stress, including delayed flowering and altered gravitropism in model crops (*Arabidopsis thaliana*)^{5,16}, dysregulation of iron uptake¹⁷, restructuring of gut bacterial communities¹⁸, and accelerated trabecular bone loss in mammalian models^{19,20}.

Here, we present a unified, comparative planetary magnetobiology framework for deep-space exploration. We extract and model the crustal magnetic environment across 36 prospective and historical landing sites—encompassing 23 lunar locations (including all 13 candidate Artemis III South Polar regions, Apollo sites, Chang’e, and Chandrayaan-3) and 13 Martian locations (including ice-rich human ISRU landing sites, active rovers, InSight ground truth, and southern anomaly belts). We synthesize these physical determinations with a systematic review of terrestrial magnetobiology literature to evaluate multi-taxonomic vulnerabilities under the triple-stressor triad (hypomagnetism, altered gravity, and cosmic radiation), establishing actionable engineering and siting recommendations for future planetary habitats.

2 Results

2.1 Global Planetary Magnetic Topology: Moon vs. Mars

Our comparative analysis reveals fundamental differences in the spatial structure, magnitude, and global organization of remnant crustal magnetization between the Moon and Mars (Table 1).

At the standard orbital mapping altitude of 30 km, the global mean lunar crustal magnetic field magnitude is $|\mathbf{B}| = 0.77 \text{ nT}$ (median = 0.56 nT), with localized peak anomalies reaching $\sim 10 \text{ nT}$ to 20 nT over the South Pole-Aitken (SPA) basin and lunar impact antipodes^{10,11}. The overall lunar magnetic landscape is characterized by widespread, homogeneous near-null magnetic field conditions across its near-side maria and polar regions.

In contrast, Mars exhibits an unprecedented planetary magnetic dichotomy at its 400 km mapping altitude (global mean $|\mathbf{B}| = 15.05 \text{ nT}$, median = 1.51 nT , peak $> 360 \text{ nT}$)¹³. The ancient Noachian crust of the southern cratered highlands (150°E to 240°E , 30°S to 75°S ; Terra Sirenum and Terra Cimmeria) harbors coherent, multipolar magnetic lineations extending thousands of kilometers, generating ground-level surface fields

exceeding 1500 nT . Conversely, the northern lowlands (Vastitas Borealis, Utopia Planitia, Arcadia Planitia) and major impact basins (Hellas, Isidis, Argyre) are severely demagnetized, exhibiting orbital fields $|\mathbf{B}_{400\text{km}}| < 2.0 \text{ nT}$ and surface fields $|\mathbf{B}_{\text{surf}}| < 25 \text{ nT}$.

2.2 Comparative Landing Site Characterization across 36 Planetary Targets

Extraction of localized vector magnetic field components across 36 landing sites establishes a continuous magnetic attenuation spectrum spanning five orders of magnitude relative to Earth’s geomagnetic baseline ($50 \mu\text{T}$) (Table 1).

2.2.1 Lunar Artemis III Candidate South Polar Regions

All 13 candidate landing regions prioritized for the Artemis III mission (84°S to 90°S) reside in an ultra-low, near-null hypomagnetic field regime. Field intensities at 30 km altitude range from 0.40 nT at Haworth crater to 0.82 nT at Peak Near Shackleton. All 13 candidate sites are classified as **low/null-field** environments ($< 1.0 \text{ nT}$), representing a $> 50,000$ -fold to $> 125,000$ -fold reduction relative to Earth’s GMF ($50 \mu\text{T}$).

2.2.2 Historical and Contemporary Lunar Landing Sites

Historical Apollo mare sites (Apollo 11, 12, 14, 15) and recent robotic landers (Chang’e 3, 5; Chandrayaan-3) exhibit near-null fields (0.10 – 1.00 nT). However, Apollo 16 in the Descartes Highlands represents a **moderate field** regime ($|\mathbf{B}_{30\text{km}}| = 4.94 \text{ nT}$), situated adjacent to a strong highland magnetic anomaly. Chang’e 4 in the Von Kármán crater on the lunar far side registers 0.72 nT at 30 km altitude within the margin of the SPA basin anomaly.

2.2.3 Martian ISRU Candidate Habitats and Rover Sites

Proposed human landing sites selected for shallow subsurface glacial ice access in the northern plains—Arcadia Planitia (39.30°N , 189.70°E) and Deuteronilus Mensae (39.10°N , 23.20°E)—reside in extreme low/null-field environments ($|\mathbf{B}_{400\text{km}}| \approx 0.70$ – 0.71 nT , ground surface $|\mathbf{B}_{\text{surf}}| \approx 5.9$ – 6.0 nT). Active and historical rover sites (Perseverance in Jezero Crater: surface $|\mathbf{B}| \approx 23.4 \text{ nT}$; Zhurong in Utopia Planitia: surface $|\mathbf{B}| \approx 6.5 \text{ nT}$; ExoMars Oxia Planum: surface $|\mathbf{B}| \approx 15.5 \text{ nT}$) are similarly situated in severely hypomagnetic regimes ($> 2,000\times$ to $> 15,000\times$ lower than Earth).

2.2.4 InSight Ground Truth and Southern Anomaly Mini-Magnetospheres

NASA’s InSight lander in Elysium Planitia (4.50°N , 135.62°E) recorded an unexpectedly high static ground surface field of $\sim 2013 \text{ nT}$ ¹⁵, demonstrating that local shallow magnetized crustal basement bodies can amplify ground-level fields by an order of magnitude over orbital predictions. In the southern highlands, candidate scientific outposts in the Terra Sirenum anomaly belt (30.00°S , 195.00°E) exhibit orbital fields of 140.07 nT and surface fields exceeding 1190 nT (regional peaks $> 1500 \text{ nT}$), forming mini-magnetospheres capable of deflecting low-energy solar wind ions¹⁴.

Table 1 — Comparative magnetic field environment across 36 lunar and Martian landing sites. Magnetic field magnitude at orbital mapping altitude ($|B_{orb}|$: 30 km for Moon, 400 km for Mars) and estimated/measured ground surface intensity ($|B_{surf}|$) derived from satellite magnetometry (Lunar Prospector, Kaguya, MGS, MAVEN) and InSight ground truth^{10,13,15,21}.

Landing Site / Target	Body / Mission	Lat (°)	Lon (°E)	$ B_{orb} $ (nT)	$ B_{surf} $ (nT)	Classification
Lunar Landing Sites (30 km Altitude Baseline)						
Faustini Rim A	Moon / Artemis III	-85.30	77.00	0.45	0.45	low/null-field
Peak Near Shackleton	Moon / Artemis III	-89.70	166.00	0.82	0.82	low/null-field
Connecting Ridge	Moon / Artemis III	-88.00	137.00	0.60	0.60	low/null-field
Connecting Ridge Ext.	Moon / Artemis III	-88.30	148.00	0.65	0.65	low/null-field
de Gerlache Rim 1	Moon / Artemis III	-88.50	290.00	0.50	0.50	low/null-field
de Gerlache Rim 2	Moon / Artemis III	-88.70	294.00	0.48	0.48	low/null-field
de Gerlache-Kocher Massif	Moon / Artemis III	-86.00	285.00	0.52	0.52	low/null-field
Haworth	Moon / Artemis III	-87.40	358.00	0.40	0.40	low/null-field
Malapert Massif	Moon / Artemis III	-86.00	0.00	0.55	0.55	low/null-field
Leibnitz Beta Plateau	Moon / Artemis III	-85.00	31.00	0.70	0.70	low/null-field
Nobile Rim 1	Moon / Artemis III	-85.40	35.00	0.62	0.62	low/null-field
Nobile Rim 2	Moon / Artemis III	-84.80	47.00	0.58	0.58	low/null-field
Amundsen Rim	Moon / Artemis III	-84.50	85.00	0.65	0.65	low/null-field
Apollo 11	Moon / Apollo	0.67	23.47	0.93	0.93	low/null-field
Apollo 12	Moon / Apollo	-3.01	336.58	1.00	1.00	low/null-field
Apollo 14	Moon / Apollo	-3.65	342.53	0.44	0.44	low/null-field
Apollo 15	Moon / Apollo	26.13	3.63	0.23	0.23	low/null-field
Apollo 16 (Highlands)	Moon / Apollo	-8.97	15.50	4.94	4.94	moderate-field
Apollo 17	Moon / Apollo	20.19	30.77	1.08	1.08	moderate-field
Chang'e 3	Moon / Chang'e	44.12	340.49	0.19	0.19	low/null-field
Chang'e 4 (SPA Basin)	Moon / Chang'e	-45.46	177.60	0.72	0.72	low/null-field
Chang'e 5	Moon / Chang'e	43.06	308.08	0.10	0.10	low/null-field
Chandrayaan-3	Moon / Chandrayaan	-69.37	32.35	0.78	0.78	low/null-field
Martian Landing Sites (400 km Altitude Baseline & Surface Modeling)						
Jezero Crater (Perseverance)	Mars / Mars 2020	18.45	77.45	2.75	23.4	low/null-field
Gale Crater (Curiosity)	Mars / MSL	-4.59	137.44	3.18	27.0	low/null-field
Elysium Planitia (InSight)	Mars / InSight	4.50	135.62	1.99	2013.0	high-field outpost
Meridiani Planum (Opportunity)	Mars / MER-B	-1.95	354.47	2.55	21.6	low/null-field
Gusev Crater (Spirit)	Mars / MER-A	-14.57	175.47	9.08	77.2	low/null-field
Vastitas Borealis (Phoenix)	Mars / Scout	68.22	234.25	0.48	4.1	low/null-field
Chryse Planitia (Viking 1)	Mars / Viking	22.48	312.05	0.59	5.0	low/null-field
Utopia Planitia (Viking 2)	Mars / Viking	47.97	134.28	0.39	3.3	low/null-field
Utopia Planitia (Zhurong)	Mars / Tianwen-1	25.07	109.92	0.76	6.5	low/null-field
Oxia Planum (ExoMars)	Mars / Rosalind Franklin	18.27	335.37	1.82	15.5	low/null-field
Arcadia Planitia (Human Candidate)	Mars / Human ISRU Base	39.30	189.70	0.71	6.0	low/null-field
Deuteronilus Mensae (Candidate)	Mars / Human ISRU Base	39.10	23.20	0.70	5.9	low/null-field
Terra Sirenum Anomaly Belt	Mars / Science Outpost	-30.00	195.00	140.07	1190.6	high-field outpost

2.3 Systematic Review of Hypomagnetic Literature

Our systematic review (Table 2) documents consistent physiological perturbations across phylogenetic groups exposed to HMF conditions ($< 5 \mu\text{T}$):

- 1. Plant Morphogenesis and Physiology:** In *Arabidopsis thaliana*, near-null magnetic fields ($< 1 \mu\text{T}$) delay floral transition via down-regulation of cryptochrome (CRY1/CRY2) and phytochrome photoperiodic signaling pathways¹⁶. HMF conditions disrupt root gravitropic curvature and meristem cell elongation by altering PIN2 polar auxin transport⁵, downregulate iron uptake transcription factors (FIT, IRT1)¹⁷, and elevate reactive oxygen species (ROS)^{1,7}.
- 2. Microbial Ecology and Magnetotaxis:** Bacterial cultures in HMFs exhibit altered cell division rates and membrane transport kinetics²², phylum-level shifts in gut microbiome community composition¹⁸, and complete loss of

magnetotaxis in magnetotactic species²³.

- 3. Animal and Human Cellular Systems:** In invertebrate models (*Drosophila*, *C. elegans*), HMF exposure extends circadian period lengths via radical pair spin state recombination modulation⁶ and alters sensory navigation²⁴. In mammalian and human cell models, HMF exposure alters neuroblastoma cell cycle transcriptomes²⁵, modifies microcirculatory capillary blood flow²⁶, and acts synergistically with mechanical unloading to accelerate trabecular bone demineralization and osteoclast resorption^{19,20}.

2.4 Multi-Planetary Biological Risk Matrix

Synthesizing physical field determinations with empirical biological sensitivity thresholds yields the multi-planetary risk assessment matrix presented in Table 3.

3 Discussion

Table 2 — Systematic synthesis of documented biological responses and biophysical mechanisms under hypomagnetic field (HMF) conditions. Summary of peer-reviewed experimental literature across plant, microbial, invertebrate, and mammalian/human systems.

Organism / Model	Condition	Biological Effect	Mechanism	Key Reference
<i>Arabidopsis thaliana</i>	Near-null (< 1 μ T)	Delayed floral transition and flowering time	Cryptochrome (CRY1/CRY2) and phytochrome signaling disruption	Agliassa et al. (2018) ¹⁶
<i>Arabidopsis thaliana</i>	Near-null (< 1 μ T)	Disrupted root directional growth and elongation	PIN2 polar auxin transport redistribution and meristem elongation	Narayana et al. (2018) ⁵
<i>Arabidopsis thaliana</i>	Hypomagnetic (< 5 μ T)	Impaired iron acquisition and uptake homeostasis	Downregulation of FIT and IRT1 transcription factors	Narayana et al. (2021) ¹⁷
<i>Arabidopsis thaliana</i>	Near-null (< 1 μ T)	Elevated reactive oxygen species (ROS) accumulation	Perturbation of redox signaling and catalase/SOD enzyme activity	Maffei (2014) ¹
Higher Plants (General)	Hypomagnetic (< 5 μ T)	Reduced photosynthetic efficiency and chlorophyll synthesis	Thylakoid membrane ultrastructural disorganization	Belyavskaya (2004) ⁷
Microorganisms (Bacteria)	Hypomagnetic (< 5 μ T)	Altered cell division rates and metabolic flux	Membrane potential modulation and altered ion channel kinetics	Creanga et al. (2009) ²²
Magnetotactic Bacteria	Near-null (< 1 μ T)	Complete loss of magnetotaxis and directional motility	Loss of passive dipole alignment of intracellular magnetosomes	Frankel et al. (1981) ²³
Gut Microbiome (Murine)	Hypomagnetic (< 5 μ T)	Restructuring of gut bacterial taxonomic diversity	Shifts in Bacteroidetes to Firmicutes phylum abundance ratios	Zhang et al. (2017) ¹⁸
<i>Drosophila melanogaster</i>	Hypomagnetic (< 5 μ T)	Circadian rhythm period lengthening and arrhythmicity	Cryptochrome-mediated radical pair recombination modulation	Yoshii et al. (2009) ⁶
<i>Drosophila melanogaster</i>	Near-null (< 1 μ T)	Embryonic developmental arrest and mortality	Mitochondrial metabolic dysfunction and ROS-induced apoptosis	Vitas et al. (2015) ²⁷
<i>Caenorhabditis elegans</i>	Hypomagnetic (< 5 μ T)	Disrupted burrowing orientation and magnetosensation	AFD sensory neuron signaling and DAF-16 FOXO pathway modulation	Vidal-Gadea et al. (2015) ²⁴
Human Neuroblastoma Cells	Hypomagnetic (< 1 μ T)	Altered cell cycle transcriptome and proliferation rate	Downregulation of cell division and DNA replication gene networks	Mo et al. (2012) ²⁵
Mammalian Model (Rat)	Hypomagnetic (< 1 μ T)	Accelerated trabecular bone demineralization and osteopenia	Synergy between HMF and hindlimb unloading; osteoblast suppression	Xu et al. (2012) ¹⁹
Mammalian Model (Rat)	Near-null (< 1 μ T)	Exacerbated musculoskeletal bone mass loss	Enhanced osteoclast differentiation and calcium excretion	Mo et al. (2014) ²⁰
Human Cardiovascular	Near-null (< 1 μ T)	Altered microcirculatory capillary blood flow velocity	Endothelial nitric oxide synthase and autonomic tone modulation	Gurfinkel et al. (2014) ²⁶
Mammalian Cells (General)	Hypomagnetic (< 1 μ T)	Altered DNA damage repair kinetics and genomic instability	Radical pair singlet-triplet spin-state interconversion dynamics	Binhi and Prato (2017) ²

3.1 Implications of Planetary Magnetic Environments for Human Habitats

The central insight established by our integrated analysis is that space biological systems and human exploration habitats deployed to the Moon and Mars will operate within two fundamentally distinct magnetic operational paradigms:

- 1. The Homogeneous Near-Null Lunar & Martian Lowland Paradigm (< 25 nT):** Primary candidate sites for initial crewed bases—specifically all 13 Artemis III South Polar regions on the Moon (< 1.0 nT) and ice-rich ISRU candidate sites on Mars such as Arcadia Planitia and Deuteronilus Mensae (< 20 nT surface)—are selected based on volatile resource availability (water ice) and solar illumination. However, our findings demonstrate that these sites represent extreme near-null hypomagnetic environments (> 2,500× to > 60,000× weaker than Earth's GMF). Bioregenerative life support systems (BLSS), agricultural growth chambers, and human crew residing in these habitats will undergo chronic, unmitigated magnetic deprivation.
- 2. The Martian Southern Highland Mini-Magnetosphere**

Outposts (> 1500 nT): In contrast, specialized scientific outposts located within the Terra Sirenum or Terra Cimmeria anomaly belts on Mars reside within intense remnant magnetic fields (> 1500 nT surface). These fields form mini-magnetospheres extending hundreds of kilometers into the Martian ionosphere¹⁴, capable of partially deflecting low-energy solar wind ions and creating localized plasma cusps. While these fields remain ~ 25× to 50× below Earth's baseline, they provide a substantially elevated baseline compared to northern lowland sites.

3.2 The Triple-Stressor Triad: Hypomagnetism, Altered Gravity, and Radiation

On extraterrestrial surfaces, magnetic field deprivation will not act in isolation. Instead, biological systems will confront a unique triple-stressor triad:

- Chronic hypomagnetic field deprivation (< 1.0 nT on Moon; < 20 nT on Martian northern plains)
- Reduced partial gravity (0.166 g on Moon; 0.38 g on Mars)
- Chronic galactic cosmic radiation (GCR) and solar particle events (SPE), delivering surface doses of ~

Table 3 — Comparative biological risk assessment matrix across lunar and Martian surface magnetic field regimes. Vulnerability of critical biological systems relative to terrestrial geomagnetic field baseline (50 μ T).

Biological System	Earth GMF (~ 50 μ T)	Mars Sirenum (> 1000 nT)	Mars InSight (~ 2000 nT)	Mars Northern Plains (< 25 nT)	Lunar Highlands (1 – 5 nT)	Lunar South Pole (< 1.0 nT)
Plant Vegetative Biomass	Baseline	Low Risk	Low Risk	High Risk	Moderate Risk	High Risk
Plant Flowering & Seed Set	Baseline	Moderate Risk	Moderate Risk	High Risk	Moderate-High Risk	High Risk
Closed-Loop Microbiome	Baseline	Low Risk	Low Risk	Moderate Risk	Low-Moderate Risk	Moderate Risk
Circadian Clocks	Baseline	Low-Moderate Risk	Low-Moderate Risk	High Risk	Moderate-High Risk	High Risk
Human Musculoskeletal	Baseline	Moderate Risk	Moderate Risk	High Risk	High Risk	High Risk
Cellular DNA / ROS Balance	Baseline	Low-Moderate Risk	Low-Moderate Risk	High Risk	Moderate-High Risk	High Risk

60 to 100 mSv/year on Moon and ~ 200 to 300 mSv/year on Mars.

Crucially, experimental evidence indicates potent biochemical synergies among these environmental factors. In mammalian hindlimb unloading models, HMF exposure significantly aggravates musculoskeletal bone loss and trabecular demineralization^{19,20}. Furthermore, HMF-induced perturbations in radical pair singlet-triplet spin-state recombination kinetics and ROS accumulation^{2,3} can impair cellular DNA repair pathways, potentially sensitizing biological systems to ionizing radiation damage.

3.3 Biophysical Mechanisms of Biological Hypomagnetic Stress

The biological sensitivity to hypomagnetic fields is mediated by two primary biophysical mechanisms:

1. **The Radical Pair Mechanism (RPM):** Flavin-based photoreceptors (cryptochromes CRY1/CRY2) rely on light-induced radical pair recombination dynamics [FAD^{•-}, TrpH^{•+}]. Removal of the background geomagnetic field alters the singlet-triplet spin interconversion rate, perturbing downstream signaling pathways governing floral transition in plants¹⁶ and circadian clock entrainment in animals⁶.
2. **Cellular Redox and Auxin Efflux Transport:** Shielding the geomagnetic field disorganizes thylakoid membrane ultrastructure⁷ and elevates mitochondrial ROS generation¹. In plant roots, this redox imbalance disrupts the polar localization of PIN2 efflux carriers, impairing directional auxin distribution and gravitropic bending⁵.

3.4 Operational Recommendations for Lunar and Martian Surface Operations

To de-risk future planetary exploration and ensure the viability of long-duration surface life support systems, we propose four core operational recommendations:

1. **Standardized Surface Magnetometry Integration:** All future precursor landers, rovers, and human surface payloads (Artemis III, CLPS, Commercial Mars missions) should integrate calibrated fluxgate magnetometers to record high-resolution ground-level magnetic fields at human operational scale.

2. **Active Magnetic Field Restoration in Growth Modules:** Agricultural and BLSS modules deployed in lunar polar craters or Martian northern lowlands should incorporate active Helmholtz coil arrays to generate artificial Earth-like static magnetic fields (~ 50 μ T), restoring normative cryptochrome signaling, auxin transport, and crop yields.
3. **Controlled Magnetic-Biology Payloads:** Early surface missions should deploy dual-chamber biological experiments (shielded near-null vs. 50 μ T artificial field) directly on the lunar and Martian surface to empirically isolate hypomagnetic effects from radiation and partial gravity.
4. **Habitat Siting Optimization:** Long-term infrastructure planning should incorporate magnetic field architecture alongside volatile access, illumination, and slope topography when evaluating permanent base siting.

4 Methods

4.1 Orbital and Surface Planetary Magnetic Field Datasets

Planetary magnetic field modeling relied on calibrated satellite and surface magnetometer observations:

1. **Lunar Datasets:** Lunar crustal magnetic field maps at 30 km altitude were obtained from the NASA Planetary Data System (PDS) archive^{10,21}, compiled from vector magnetometer data collected by the Lunar Prospector (LP-MAG) and SELENE (Kaguya) missions. Equatorial and mid-latitude coverage (65°S to 65°N) at ~ 0.5° resolution was integrated with high-resolution polar models (60° to 90°S)²¹ and surface anomaly models¹¹.
2. **Martian Datasets:** Martian crustal magnetic field determinations were derived from the continuous spherical harmonic model of Langlais et al.¹³ (DOI: 10.1029/2019JE005979), combining vector magnetometer observations from the Mars Global Surveyor (MGS MAG/ER) at 400 km mapping orbit and periapsis passes (~ 130 km) with low-altitude MAVEN nightside observations. Ground-level surface field magnitudes ($|\mathbf{B}_{\text{surf}}|$) were estimated via downward-continuation scaling and surface magnetic field extrapolation validated against direct in situ surface magnetometry from the NASA InSight fluxgate magnetometer¹⁵.

Total magnetic field intensity ($|\mathbf{B}|$) was calculated from orthogonal spatial components:

$$|\mathbf{B}| = \sqrt{B_r^2 + B_\theta^2 + B_\phi^2} \quad (1)$$

where B_r is radial, B_θ is colatitudinal (southward), and B_ϕ is azimuthal (eastward).

4.2 Landing Site Extraction and Bilinear Interpolation

A unified database of 36 planetary landing coordinates was constructed, encompassing 23 lunar sites (13 Artemis III candidate South Polar regions²⁸, Apollo 11, 12, 14, 15, 16, 17, Chang'e 3, 4, 5, and Chandrayaan-3) and 13 Martian sites (Arcadia Planitia, Deuteronilus Mensae, Terra Sirenum, Perseverance, Curiosity, InSight, Spirit, Opportunity, Phoenix, Viking 1/2, Zhurong, ExoMars). Local vector magnetic field components were extracted using 2D bilinear spatial interpolation:

$$f(\lambda, \phi) \approx \sum_{i=1}^2 \sum_{j=1}^2 w_{ij} f(\lambda_i, \phi_j) \quad (2)$$

where w_{ij} represent inverse-distance spatial weighting factors across latitude λ and longitude ϕ .

4.3 Systematic Review Protocol

A systematic review of hypomagnetic field biological effects was conducted across PubMed, Web of Science, and NASA Technical Reports (2000–2026). Search queries combined terms “hypomagnetic field”, “near-null magnetic field”, and “zero magnetic field” with target biological endpoints (“Arabidopsis”, “cryptochrome”, “auxin”, “ROS”, “microbiome”, “osteoclast”, “bone demineralization”, “radical pair”, “DNA repair”). Peer-reviewed experimental studies utilizing Mu-metal or active Helmholtz coil shielding achieving field attenuation below 5 μ T were prioritized for quantitative synthesis.

4.4 Software and FAIR Data Availability

Data processing pipelines were executed in Python using `numpy`, `pandas`, and `scipy`. LaTeX tables were generated automatically using custom scripts. The complete codebase, processed datasets, and publication source files are archived on Zenodo (DOI assigned upon publication) and hosted at <https://github.com/dr-richard-barker/lunar-mars-magnetic-biology>.

5 Conclusion

This study establishes a unified, comparative planetary magnetobiology framework for deep-space exploration, quantitatively mapping the remnant magnetic environments across 36 landing sites on the Moon and Mars and evaluating their biological implications. Our analysis demonstrates that all 13 candidate Artemis III South Polar landing sites on the Moon (< 1.0 nT) and ice-rich human habitat candidate sites in the Martian northern plains (< 20 nT) represent severe, near-null hypomagnetic field environments ($> 2,500$ -fold to $> 60,000$ -fold weaker

than Earth's geomagnetic baseline). In contrast, Martian southern highland anomalies form localized mini-magnetospheres exceeding 1500 nT at ground level. Synthesizing empirical magnetobiology literature indicates that unmitigated hypomagnetic exposure poses distinct developmental hazards to crop plants, alters closed-loop habitat microbiomes, and synergistically accelerates musculoskeletal bone loss and cellular DNA repair perturbations in astronauts. We conclude that planetary magnetic field architecture is an essential environmental factor in space exploration, and we recommend integrating active Helmholtz coil magnetic restoration into extraterrestrial growth modules alongside standardized surface magnetometry on precursor landers.

Data Availability

The planetary crustal magnetic field datasets analyzed in this study are available through the NASA Planetary Data System (PDS) Planetary Plasma Interactions (PPI) and Geosciences Nodes under DOIs [10.17189/1520494](https://doi.org/10.17189/1520494), [10.17189/rk57-g992](https://doi.org/10.17189/rk57-g992), and [10.1029/2019JE005979](https://doi.org/10.1029/2019JE005979). Processed landing site tables, comparative systematic review datasets, and interactive WebGL visualization assets are archived in Zenodo (DOI will be assigned upon publication) and hosted at <https://github.com/dr-richard-barker/lunar-mars-magnetic-biology>.

Code Availability

All Python analysis scripts, processing pipelines, figure generation modules, and manuscript source files are open source under the MIT license at <https://github.com/dr-richard-barker/lunar-mars-magnetic-biology>.

Acknowledgements

The authors acknowledge the science and instrument teams of the Lunar Prospector, SELENE (Kaguya), Mars Global Surveyor (MGS) MAG/ER, MAVEN, and InSight missions for collecting and archiving planetary magnetic field observations. This work was supported by space biology and bioengineering research initiatives at Purdue University.

Author Contributions

R.B. and D.M.P. conceived the research and experimental framework. R.B., A.K.S., M.D., and K.B. developed the data processing pipeline, conducted the magnetobiology literature review, and prepared the figures and tables. C.S.B. contributed planetary magnetism modeling, lunar/Martian paleomagnetism expertise, and dataset interpretation. R.B., C.S.B., and D.M.P. drafted the manuscript and formulated the comparative biological risk matrix. All authors reviewed, edited, and approved the final manuscript.

Competing Interests

The authors declare no competing financial or non-financial interests.

References

1. M. E. Maffei. Magnetic field effects on plant growth, development, and evolution. *Frontiers in Plant Science*, 5: 445, 2014. doi: 10.3389/fpls.2014.00445.
2. V. N. Binh and F. S. Prato. Biological effects of the hypomagnetic field: An analytical review of experiments and theories. *PLoS ONE*, 12:e0179340, 2017. doi: 10.1371/journal.pone.0179340.
3. P. J. Hore and H. Mouritsen. The radical-pair mechanism of magnetoreception. *Annual Review of Biophysics*, 45:299–344, 2016. doi: 10.1146/annurev-biophys-032116-094545.
4. T. Ritz, S. Adem, and K. Schulten. A model for photoreceptor-based magnetoreception in birds. *Biophysical Journal*, 78:707–718, 2000. doi: 10.1016/S0006-3495(00)76629-X.
5. R. Narayana, J. Fliegmann, I. Paponov, and M. E. Maffei. Reduction of geomagnetic field (GMF) to near null magnetic field (NNMF) affects *Arabidopsis thaliana* root growth and morphology by altering polar auxin transport. *Plant Physiology and Biochemistry*, 130:64–71, 2018. doi: 10.1016/j.plaphy.2018.06.028.
6. T. Yoshii, M. Ahmad, and C. Helfrich-Förster. Cryptochrome mediates light-dependent magnetosensitivity of *Drosophila*'s circadian clock. *PLoS Biology*, 7:e1000086, 2009. doi: 10.1371/journal.pbio.1000086.
7. N. A. Belyavskaya. Biological effects due to weak magnetic field on plants. *Advances in Space Research*, 34:1566–1574, 2004. doi: 10.1016/j.asr.2004.01.021.
8. M. H. Acuña, J. E. P. Connerney, N. F. Ness, R. P. Lin, D. Mitchell, C. W. Carlson, J. McFadden, K. A. Anderson, H. Rème, C. Mazelle, D. Vignes, P. Wasilewski, and P. Cloutier. Global distribution of crustal magnetization discovered by the Mars Global Surveyor MAG/ER experiment. *Science*, 284:790–793, 1999. doi: 10.1126/science.284.5415.790.
9. M. A. Wieczorek, B. L. Jolliff, A. Khan, M. E. Pritchard, B. P. Weiss, J. G. Williams, and M. T. Zuber. The constitution and structure of the lunar interior. *Reviews in Mineralogy and Geochemistry*, 60:221–364, 2006. doi: 10.2138/rmg.2006.60.3.
10. L. L. Hood, C. B. Torres, J. S. Oliveira, M. A. Wieczorek, and S. T. Steward. Lunar crustal magnetic field map, 30 km altitude. NASA Planetary Data System, 2020.
11. H. Tsunakawa, F. Takahashi, H. Shimizu, H. Shibuya, and M. Matsushima. Regional mapping of the lunar magnetic anomalies at the surface: Method and its application to strong and weak magnetic anomaly regions. *Journal of Geophysical Research: Planets*, 120:1160–1185, 2015. doi: 10.1002/2014JE004785.
12. J. E. P. Connerney, M. H. Acuña, N. F. Ness, G. Kletetschka, D. L. Mitchell, R. P. Lin, H. Rème, and C. Mazelle. Tectonic implications of Mars crustal magnetism. *Proceedings of the National Academy of Sciences*, 102(42):14970–14975, 2005. doi: 10.1073/pnas.0507469102.
13. B. Langlais, E. Thébault, B. Sanchez-Cano, J. E. P. Connerney, A. Mittelholz, and C. L. Johnson. A new model of the crustal magnetic field of mars using MAVEN and Mars Global Surveyor data. *Journal of Geophysical Research: Planets*, 124(6):1542–1569, 2019. doi: 10.1029/2019JE005979.
14. D. L. Mitchell, R. P. Lin, C. Mazelle, H. Rème, P. A. Cloutier, J. E. P. Connerney, M. H. Acuña, and N. F. Ness. Probing Mars' crustal magnetic field and ionosphere with the MGS electron reflectometer. *Journal of Geophysical Research: Planets*, 106:23419–23427, 2001. doi: 10.1029/2000JE001435.
15. C. L. Johnson, A. Mittelholz, B. Langlais, C. T. Russell, V. Ansan, D. Banfield, P. J. Chi, M. O. Fillingim, F. Forget, A. Gao, S. Goossens, M. Grott, J. Halekas, E. Hauber, D. Jault, B. Knapmeyer-Endrun, M. Lemmon, N. Mueller, M. Panning, S. E. Smrekar, A. Spiga, R. Weber, M. A. Wieczorek, and W. B. Banerdt. Crustal and time-varying magnetic fields at the InSight landing site on Mars. *Nature Geoscience*, 13:199–204, 2020. doi: 10.1038/s41561-020-0536-x.
16. C. Agliassa, R. Narayana, J. M. Christie, and M. E. Maffei. Geomagnetic field impacts on cryptochrome and phytochrome signaling in *Arabidopsis thaliana*. *Journal of Photochemistry and Photobiology B: Biology*, 185:32–40, 2018. doi: 10.1016/j.jphotobiol.2018.05.027.
17. R. Narayana, J. Fliegmann, and M. E. Maffei. Near-null magnetic field alters iron homeostasis gene expression in *Arabidopsis thaliana*. *Environmental and Experimental Botany*, 188:104527, 2021. doi: 10.1016/j.envexpbot.2021.104527.
18. B. Zhang, Y. Wang, R. Zha, X. Sun, C. Liu, and W. Mo. Hypomagnetic field induces structural and functional alterations of the gut microbiome in mice. *Frontiers in Microbiology*, 8:2340, 2017. doi: 10.3389/fmicb.2017.02340.
19. C. Xu, X. Yin, Y. Wu, and J. Chong. A hypomagnetic field aggravates bone loss induced by hindlimb unloading in rat. *Advances in Space Research*, 50:414–418, 2012. doi: 10.1016/j.asr.2012.04.020.

20. W. Mo, Y. Liu, and R. He. A hypomagnetic field aggravates bone loss induced by hindlimb unloading in rat. *PLoS ONE*, 9:e105604, 2014. doi: 10.1371/journal.pone.0105604.
21. L. L. Hood, I. Bryant, and J. van der Leeuw. Lunar polar crustal magnetic field maps, 30 km altitude. NASA Planetary Data System, 2022.
22. D. E. Creanga, C. Nadejde, and M. Lica. Low magnetic field effects on some bacteria. *Romanian Journal of Physics*, 54:215–223, 2009.
23. R. B. Frankel, R. P. Blakemore, F. F. Torres de Araujo, D. M. Esquivel, and J. Danon. Magnetotactic bacteria at the geomagnetic equator. *Science*, 212:1269–1270, 1981. doi: 10.1126/science.212.4500.1269.
24. A. Vidal-Gadea, K. Ward, C. Beron, N. Ghorashian, S. Gokce, J. Russell, N. Truong, A. Parikh, O. Ghaed, A. Ben-Yakar, and J. Pierce-Shimomura. Magnetosensitive neurons mediate geomagnetic orientation in *Caenorhabditis elegans*. *eLife*, 4:e07493, 2015. doi: 10.7554/eLife.07493.
25. W. Mo, Y. Liu, P. F. Bartlett, and R. He. Transcriptome profile of human neuroblastoma cells in the hypomagnetic field. *Science China Life Sciences*, 55:878–886, 2012. doi: 10.1007/s11427-012-4394-x.
26. Y. I. Gurfinkel, O. Y. At'kov, A. L. Vasin, and M. L. Sasonko. Effect of zero magnetic field on cardiovascular system in humans. *Bioelectromagnetics*, 35:237–244, 2014. doi: 10.1002/bem.21833.
27. M. Vitas, M. Vitas, and M. Čemažar. The effect of hypomagnetic field on early development of *Drosophila melanogaster*. *Bioelectromagnetics*, 36:483–491, 2015. doi: 10.1002/bem.21927.
28. NASA. Artemis iii candidate landing regions. Technical report, National Aeronautics and Space Administration, 2022. URL <https://www.nasa.gov/specials/artemis-iii/>.